Joint CDF

Let X and Y be random variables (could be either discrete or continuous). Joint Cumulative Distribution Function (CDF) is defined as:

$$F_{XY}(x, y) = P(X \leq x, Y \leq y).$$

Joint pmf

Given X and Y are **discrete** random variables defined on the same sample space and that they take on values $x_1, x_2, \cdots$ and $y_1, y_2, \cdots$ respectively, joint probability mass function (pmf) $p(x, y)$ is

$$p_{XY}(x_i, y_i) = P\{X = x_i, Y = y_i\}.$$

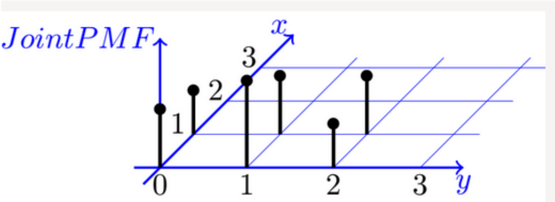
From here, the respective **marginal pmf** of X and Y are:

$$p_X(x_i) = \sum_j \underbrace{p_{XY}(x_i, y_j)}_{\text{Sum over all the } y_j} \quad \text{and } p_Y(y_j) = \sum_i \underbrace{p_{XY}(x_i, y_j)}_{\text{Sum over all the } x_i}.$$

https://www.probabilitycourse.com/chapter5/5_1_1_joint_pmf.php

Consider two random variables X and Y with joint pmf given

	$Y=0$	$Y=1$	$Y=2$
$X=0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X=1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$



Probability of event pair: 'height' of joint pmf.

Independence of random variables

R.v. X and Y are **independent** if and only if for any real numbers a, b,

$$P(X \leq a, Y \leq b) = P\{X \leq a\}P\{Y \leq b\} \text{ or } F_{XY}(a,b) = F_X(a)F_Y(b).$$

Or (in the discrete case)

$$p_{XY}(a,b) = p_X(a)p_Y(b) \quad \forall a, b \in \Omega.$$

Sum of Independent Random Variables

Suppose that X, Y are **independent discrete** random variables. Then $Z = X + Y$ is a discrete random variable with

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x).$$

- Sum of independent Poisson r.v. is Poisson.
- Sum of independent Binomial r.v. is Binomial.

(Discrete) Conditional Probability Of $X = x_i$ given $Y = Y_j$

$$p_{X|Y}(x_i|y_j)$$
$$= P\{X = x_i | Y = y_j\} = \frac{P\{X = x_i \text{ and } Y = y_j\}}{P(Y = y_j}= \frac{p_{XY}(x_i, y_j)}{p_Y(y_j)}.$$

assuming that $p_Y(y_j) > 0$.

- $\sum_i p_{X|Y}(x_i|y_j) = 1$.
- If X and Y are independent, then $p_{X|Y}(x_i|y_j) = p_X(x_i)$.

Total Probability

- Multiplicative rule: $p_{XY}(x, y) = p_{X|Y}(x|y)p_Y(y)$.
- Law of total probability: $p_X(x) = \sum_y p_{X|Y}(x|y)p_Y(y)$.

Joint CDF and pdf

Given X and Y are jointly **continuous**, the joint pdf of X and Y is: $f(x, y)$ where

- $f(x, y) \geq 0$,
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$.

The joint CDF of X and Y is:

$$F(x, y) = P\{X \leq x, Y \leq y\} = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, dv\, du.$$

Note:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$

Marginal pdf

The respective **marginal pdf** of X and Y are:

$$f_X(x) = \underbrace{\int_{-\infty}^{\infty} f(x,y)\,dy}_{\text{Integrate over all the } y} \quad \text{and} \quad f_Y(y) = \underbrace{\int_{-\infty}^{\infty} f(x,y)\,dx}_{\text{Integrate over all the } x}.$$

Note: State the domain of the marginal pdf.

Proposition: Independence of random variables

X and Y are independent if and only if their joint probability density/mass function can be expressed as

$$f_{XY}(x,y) = h(x)g(y), -\infty < x, y < \infty.$$

where $h(x)$ **depends only on** x and $g(y)$ **depends only on** y.
Remark: To show that X and Y are not independent, it suffices to show that

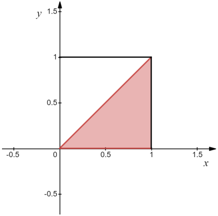
$$f_{XY}(x,y) \neq f_X(x) \cdot f_Y(y).$$

The joint density of X and Y is given by

$$f(x,y) = \frac{12}{7}(x^2 + xy), \quad 0 \le x \le 1, 0 \le y \le 1.$$

Find $P\{X > Y\}$.

[Solution:] Integrate f over the set $A := \{f(x,y) \mid 0 \le y \le x \le 1\}$.



$$P(X > Y) = \int_0^1 \int_y^1 \frac{12}{7}(x^2 + xy)\, dx\, dy = \frac{12}{7} \int_0^1 \left[\frac{x^3}{3} + \frac{x^2}{2} y \right]_{x=y}^{x=1} dy$$

$$= \frac{12}{7} \int_0^1 \frac{1}{3} + \frac{1}{2} y - \frac{5}{6} y^3\, dy = \frac{9}{14}.$$

Sum of independent continuous random variables (Convolution)

When X and Y are **independent** continuous random variables, $Z = X + Y$ has the probability density function.

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx.$$

Let X and Y be the independent uniform random variables on $(0, 1)$. Find the density of $X + Y$.

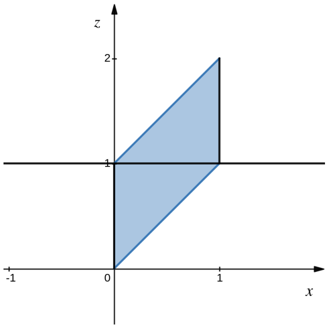
[Solution:] Note that:

$$f_X(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad f_Y(z - x) = \begin{cases} 1 & \text{if } 0 \leq z - x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Integrand: $f_X(x) f_Y(z - x) = 1$ only when $0 \leq x \leq 1$ AND $0 \leq z - x \leq 1$.

Rearrange the inequality $0 \leq z - x \leq 1$ into $z - 1 \leq x \leq z$.

Sketch the described region of $0 \leq x \leq 1$ and $z - 1 \leq x \leq z$.



- For $z \leq 1$,

  we have $0 \leq x \leq z$ for $0 \leq z \leq 1$. $f_Z(x) = \displaystyle\int_0^z 1\,dx = z$.

- For $1 \leq z \leq 2$,

  we have $z - 1 \leq x \leq 1$ for $1 \leq z \leq 2$. $f_Z(x) = \displaystyle\int_{z-1}^1 1\,dx = 2 - z$.